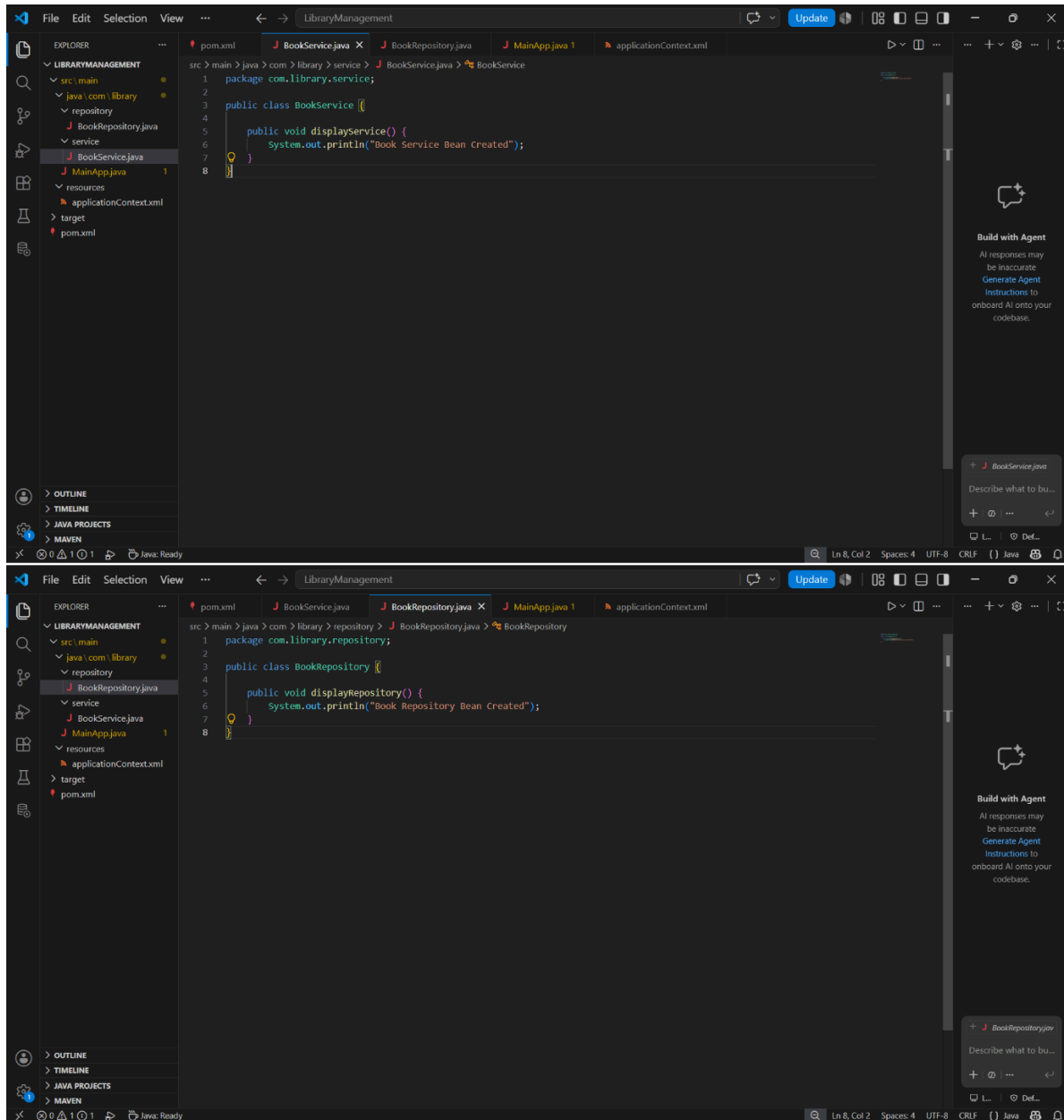


DIGITAL NURTURE 5.0 DEEPSKILLING

WEEK 2 – HANDS-ON

Spring Core and Maven

Exercise 1: Configuring a Basic Spring Application



File Edit Selection View ... LibraryManagement Update

EXPLORER

LIBRARYMANAGEMENT

src/main

java.com.library

repository

BookRepository.java

service

BookService.java

MainApp.java

resources

applicationContext.xml

target

pom.xml

src > main > java > com > library > J MainApp.java > ...

3 import org.springframework.context.ApplicationContext;

4 import org.springframework.context.support.ClassPathXmlApplicationContext;

5

6 import com.library.service.BookService;

7 import com.library.repository.BookRepository;

8

9 public class MainApp {

10

11 Run | Debug

12 public static void main(String[] args) {

13

14 ClassPathXmlApplicationContext context = new ClassPathXmlApplicationContext(configLocation: "applicationContext.xml");

15

16 BookService service =

17 (BookService) context.getBean(name: "bookService");

18

19 BookRepository repository =

20 (BookRepository) context.getBean(name: "bookRepository");

21

22 service.displayService();

23 repository.displayRepository();

24 context.close();

25 }

Build with Agent

AI responses may be inaccurate

Generate Agent

Instructions to onboard AI onto your codebase.

+ J MainApp.java

Describe what to bu...

+ ...

Del...

Ln 3, Col 26 Spaces: 4 UTF-8 CRLF Java Ready

File Edit Selection View ... LibraryManagement Update

EXPLORER

LIBRARYMANAGEMENT

src/main

java.com.library

repository

BookRepository.java

service

BookService.java

MainApp.java

resources

applicationContext.xml

target

pom.xml

src > main > resources > applicationContext.xml

1 <?xml version="1.0" encoding="UTF-8"?>

2

3 <beans xmlns="http://www.springframework.org/schema/beans"

4 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

5 xsi:schemaLocation="

6 http://www.springframework.org/schema/beans

7 https://www.springframework.org/schema/beans/spring-beans.xsd">

8

9 <bean id="bookService"

10 class="com.library.service.BookService"/>

11

12 <bean id="bookRepository"

13 class="com.library.repository.BookRepository"/>

14

15 </beans>

16

Build with Agent

AI responses may be inaccurate

Generate Agent

Instructions to onboard AI onto your codebase.

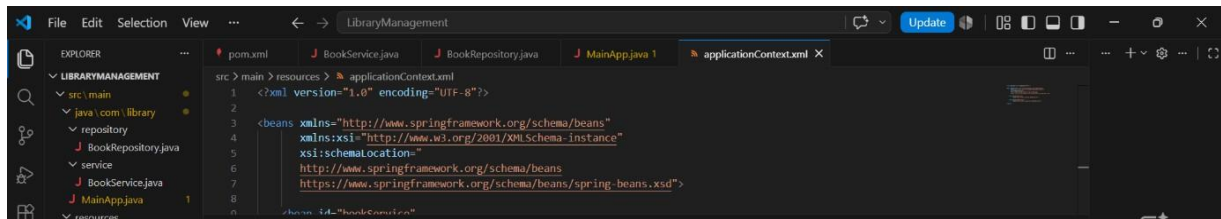
+ applicationContext

Describe what to bu...

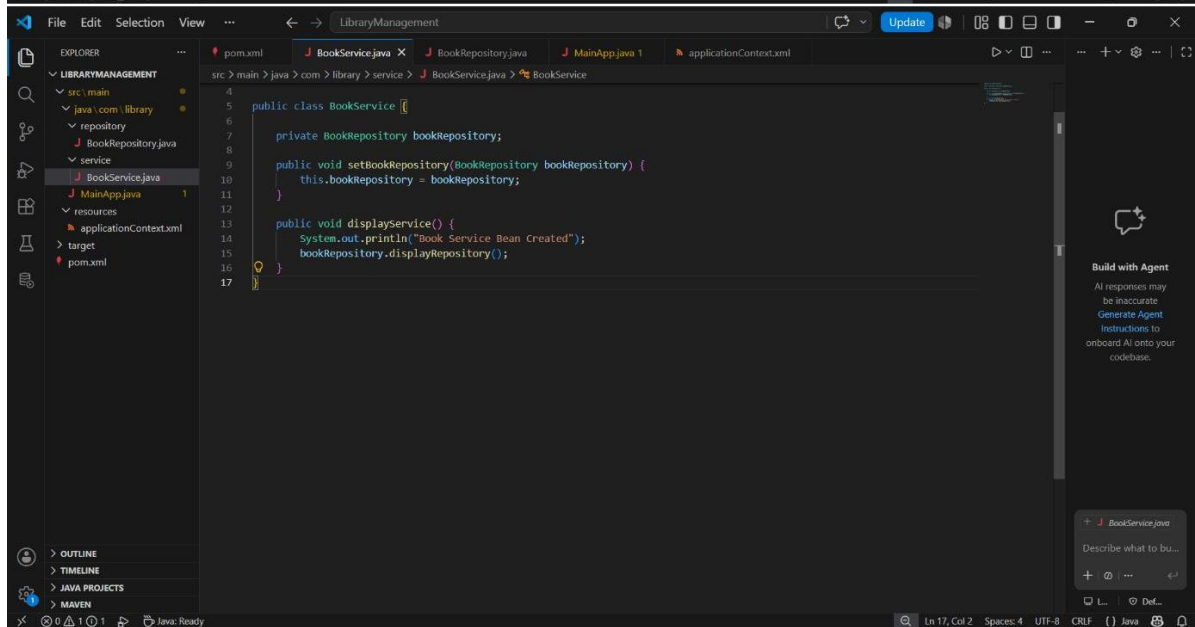
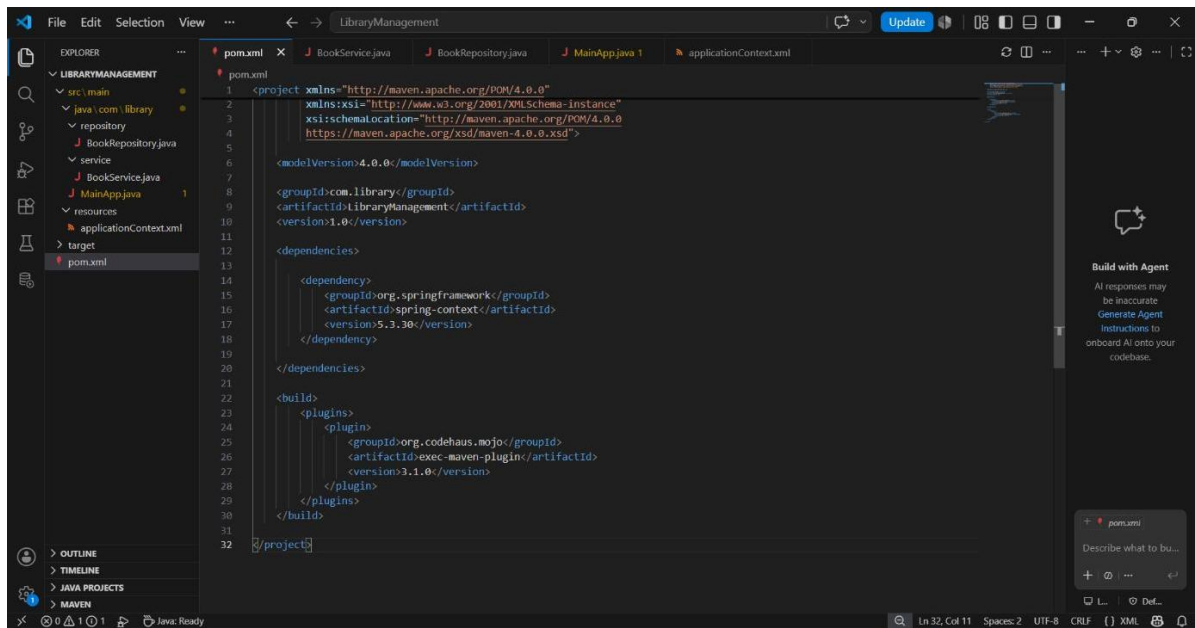
+ ...

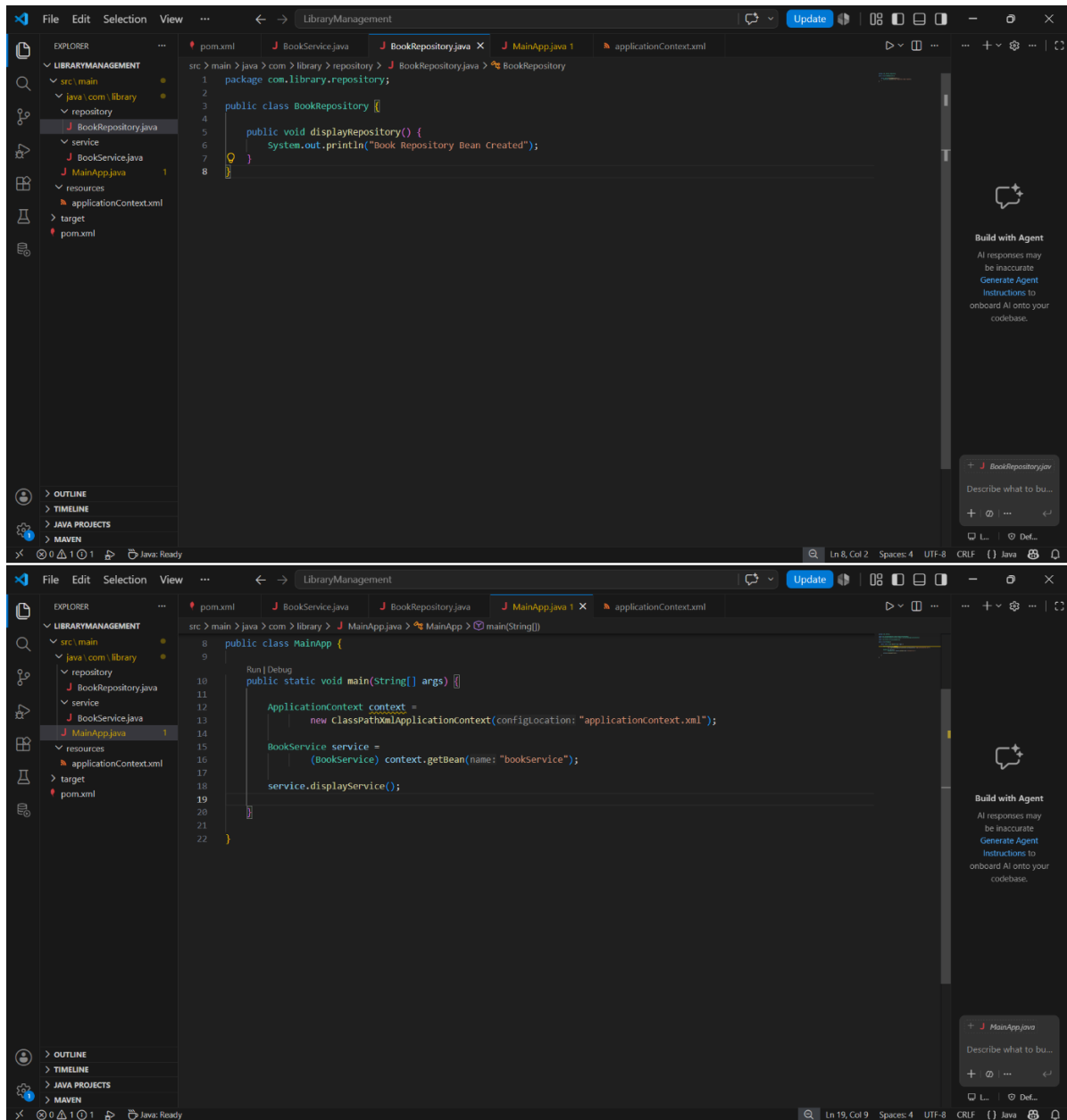
Del...

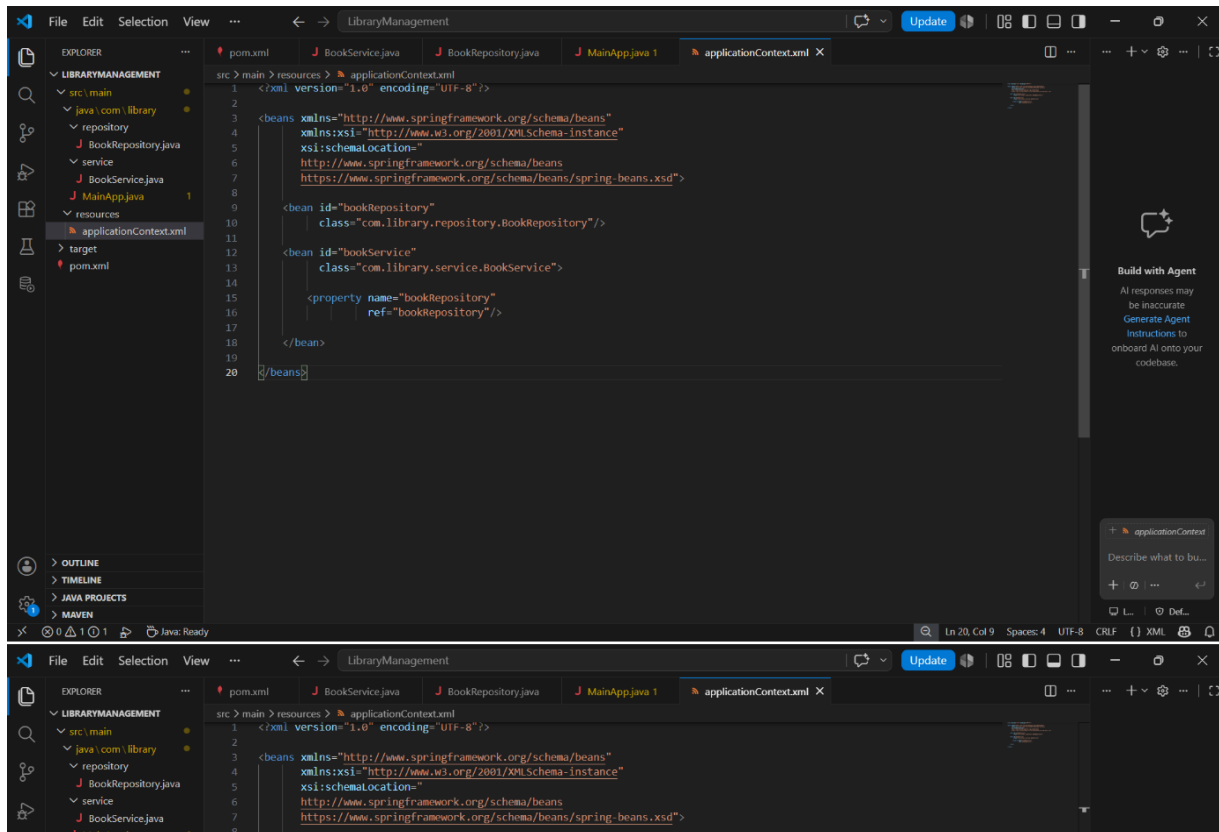
Ln 16, Col 1 Spaces: 4 UTF-8 CRLF XML Java Ready



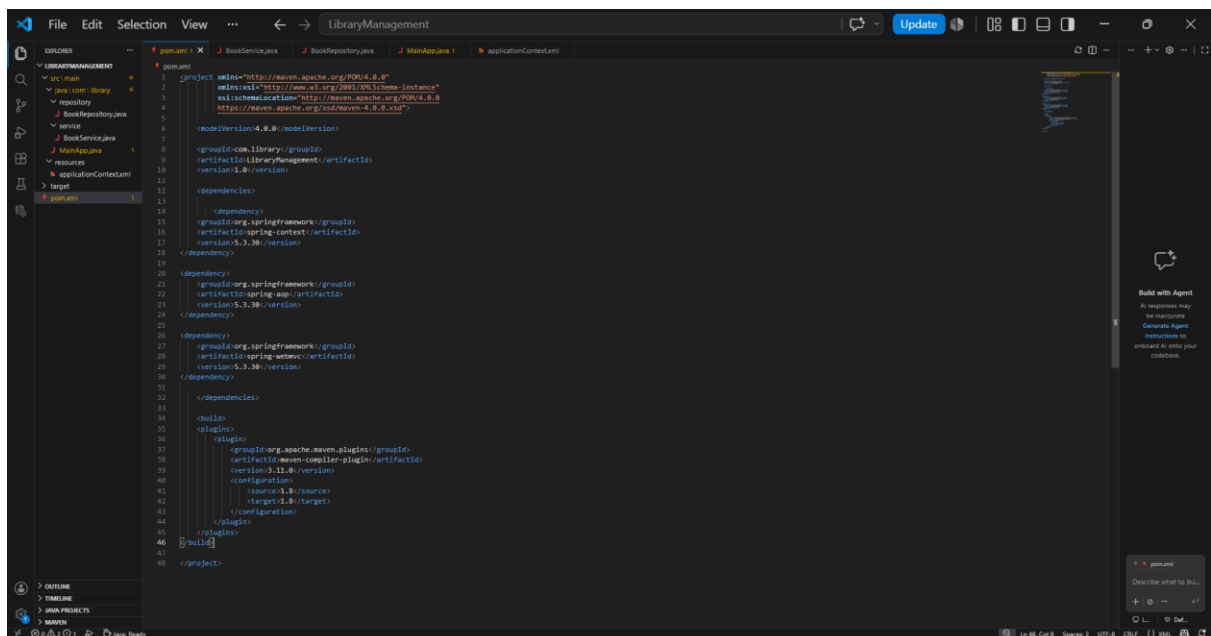
Exercise 2: Implementing Dependency Injection







Exercise 4: Creating and Configuring a Maven Project



Spring Data JPA with Spring Boot, Hibernate

Spring Data JPA - Quick Example

application.properties

Spring Framework and application log

logging.level.org.springframework=info

logging.level.com.cognizant=debug

Hibernate logs

logging.level.org.hibernate.SQL=trace

logging.level.org.hibernate.type.descriptor.sql=trace

Log pattern

logging.pattern.console=%d{dd-MM-yy} %d{HH:mm:ss.SSS} %-20.20thread %5p %-25.25logger{25}
%25M %4L %m%n

Database configuration

spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver

spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn

spring.datasource.username=root

spring.datasource.password=root

Hibernate configuration

spring.jpa.hibernate.ddl-auto=validate

spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQL5Dialect

Country.java

package com.cognizant.ormlearn.model;

import javax.persistence.Column;

import javax.persistence.Entity;

import javax.persistence.Id;

import javax.persistence.Table;

@Entity

@Table(name = "country")

public class Country {

 @Id

```

@Column(name = "co_code")
private String code;

@Column(name = "co_name")
private String name;

public String getCode() {
    return code;
}

public void setCode(String code) {
    this.code = code;
}

public String getName() {
    return name;
}

public void setName(String name) {
    this.name = name;
}

@Override
public String toString() {
    return "Country [code=" + code + ", name=" + name + "]";
}
}

```

CountryRepository.java

```

package com.cognizant.ormlearn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

import com.cognizant.ormlearn.model.Country;

@Repository

public interface CountryRepository extends JpaRepository<Country, String> {
}

```

CountryService.java

```

package com.cognizant.ormlearn.service;

import java.util.List;

import javax.transaction.Transactional;

import org.springframework.beans.factory.annotation.Autowired;

```

```

import org.springframework.stereotype.Service;
import com.cognizant.ormlearn.model.Country;
import com.cognizant.ormlearn.repository.CountryRepository;

@Service

public class CountryService {

    @Autowired

    private CountryRepository countryRepository;

    @Transactional

    public List<Country> getAllCountries() {

        return countryRepository.findAll();

    }

}

```

OrmLearnApplication.java

```

package com.cognizant.ormlearn;

import java.util.List;

import org.slf4j.Logger;

import org.slf4j.LoggerFactory;

import org.springframework.boot.SpringApplication;

import org.springframework.boot.autoconfigure.SpringBootApplication;

import org.springframework.context.ApplicationContext;

import com.cognizant.ormlearn.model.Country;

import com.cognizant.ormlearn.service.CountryService;

@SpringBootApplication

public class OrmLearnApplication {

    private static final Logger LOGGER =

        LoggerFactory.getLogger(OrmLearnApplication.class);

    private static CountryService countryService;

    public static void main(String[] args) {

        ApplicationContext context =

            SpringApplication.run(OrmLearnApplication.class, args);

        countryService = context.getBean(CountryService.class);

        LOGGER.info("Inside main");

        testGetAllCountries();
    }

}

```



```

    }

    private static void testGetAllCountries() {

        LOGGER.info("Start");

        List<Country> countries =

            countryService.getAllCountries();

        LOGGER.debug("countries={}", countries);

        LOGGER.info("End");

    }

}

```

pom.xml

```

<dependencies>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-data-jpa</artifactId>
    </dependency>
    <dependency>
        <groupId>mysql</groupId>
        <artifactId>mysql-connector-java</artifactId>
        <scope>runtime</scope>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-devtools</artifactId>
        <scope>runtime</scope>
        <optional>true</optional>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
    </dependency>
</dependencies>

```

MySQL Script

```
create schema ormlearn;
```

```
use ormlearn;
```

```
create table country(  
    co_code varchar(2) primary key,  
    co_name varchar(50)  
);
```

```
insert into country values ('IN','India');
```

```
insert into country values ('US','United States of America');
```

Maven Command

```
mvn clean package
```

Run Application

```
mvn spring-boot:run
```

Log:

Inside main

Start

```
countries=[Country [code=IN, name=India], Country [code=US, name=United States of America]]
```

End